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Project Proposal

HunarHub: A Web-Based Peer-to-Peer Skill Sharing and Learning Platform

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1. Project Title

HunarHub: A Web-Based Peer-to-Peer Skill Sharing and Learning Platform

2. Introduction

HunarHub is a web application where Learners and professionals can teach what they know and learn what they need. "Hunar" means skill in Urdu, and the idea is simple: a hub where people with skills and people who want those skills can find each other, talk, and set up learning sessions without leaving the platform.

Every member builds a profile with a photo, a short introduction, education, languages, and an optional intro video. Someone who wants to teach adds their skills and courses, each with a level, an outline, and a portfolio of past work. Someone who wants to learn lists the skills they are looking for and searches for Instructors. The learner sends a connection request, and once the instructor accepts, the two can chat, share files and voice messages, book a session, and meet online.

Three features set up HunarHub apart from an ordinary tutoring or freelancing site. **Skill Swap** shows when two members can teach each other. **Smart Match** suggests the most suitable Instructors for a learner. The **Hunar Score** shows how complete and trusted a profile is. Instructors can also create groups for their Learners and hold group meetings, and an admin panel lets the platform handle reports and keep the community safe.

Through this project we will put into practice web development, database design, real-time communication, interface design, and web application security.

3. Problem Statement

A Learner who wants to learn from a peer, or a professional who is happy to teach, has no single trusted place to meet. Most people end up using social media groups, messaging apps, and a separate video tool. This causes a few clear problems:

- Finding an instructor for one specific skill is hard, and there is no reliable way to judge their experience or teaching quality.
- Learners and Instructors jump between several apps for chatting, scheduling, sharing files, and meeting.
- Without structured profiles, course outlines, and reviews, trust depends only on personal contacts.
- Freelancing platforms are made for paid projects, not for Learners who want to exchange skills with each other.
- An Instructor with many Learners struggles to keep communication organized.
- Users have very little control over privacy, and blocking or reporting to someone who behaves badly is difficult.

What is needed is one secure web platform that brings profiles, search, chat, scheduling, reviews, group learning, and online meetings together.

4. Objectives

The main objectives of the project are to:

1. Build a web platform where every member can teach and learn.
2. Let members create professional profiles with skills, course outlines, and portfolios.
3. Provide search and filters so learners can find suitable Instructors quickly.
4. Recommend Instructors through Smart Match and highlight Skill Swap opportunities.
5. Require a connection request to be accepted before two members can message each other.
6. Provide a messenger that supports text, files, photos, links, and voice messages.
7. Let learners book sessions and leave ratings and reviews afterwards.
8. Support groups so an instructor can talk to many Learners at once.
9. Support online meetings so members do not need another platform.
10. Provide blocking, reporting, and admin moderation to keep users safe.
11. Protect accounts with hashed passwords, email verification codes, and request checks.
12. Make the design responsive for both mobile and desktop browsers and keep the back end ready for a mobile application later.

5. Scope of the System

Learner side

- Register, log in, and reset a forgotten password with a six-digit code sent by email
- Create and edit a profile
- List the skills they want to learn
- Search Instructors by skill, category, level, and price
- See recommended Instructors through Smart Match
- Send connection requests and receive notifications
- Chat with connected members, including files and voice messages
- Book a session inside the instructor's weekly availability
- Join online sessions and meetings
- Rate and review Instructors
- Post learning requests on the Learning Board and receive offers
- Join groups created by Instructors
- Block and report members

Instructure side

- Create a teaching profile with skills, course outlines, and a portfolio
- Add an intro video and weekly availability
- Accept or reject connection requests and session requests
- Mark sessions as completed
- Send offers to learning requests on the Learning Board
- Create groups, add connected Learners, and chat with the whole group
- Start group meetings and video calls
- View ratings, reviews, and the Hunar Score

Admin side

- Log in to a protected admin panel
- Review reports submitted by users
- Search members and suspend or restore accounts
- View statistics about members and reports

Not included in this version

- Online payments and a credit-based payment system (see Future Work)
- A mobile application (planned after the web application goes live, see Future Work)
- Our own video meeting servers (meetings use the Jitsi Meet service)

6. System Features

User authentication and security. Passwords are stored as hashes; repeated wrong logins are limited, password reset uses a time-limited code sent by email, and requests coming from other websites are blocked.

Profiles and portfolios. Each member has a photo, headline, bio, education, languages, skills with course outlines, portfolio projects, and an optional YouTube intro video.

Hunar Score. A score out of 100 based on profile completeness, skills, portfolio, and reviews, shown together with tips on how to improve it.

Search and filter. Members can search by skill, category, level, and maximum price, and sort results by rating, price, score, or newest.

Smart Match. Instructors are ranked using the learner's wanted skills, Skill Swap, ratings, availability, and location. The system also shows the reason behind each match.

Skill Swap. When two members can teach each other, both see a Perfect Swap badge.

Connections. One member sends a request, and the other must accept it before they can message. Pending and accepted requests show up as notifications.

Messenger. Text, photos, documents, links, and voice messages, with seen status, message editing, unsend, and delete for me.

Weekly availability and sessions. Instructors publish weekly time slots, learners book inside them, and Instructors confirm or decline.

Ratings and reviews. After a completed session, a learner rates the instructor from one to five stars and can write a comment.

Learning Board. Learners post what they want to learn, and Instructors send offers. Accepting an offer connects both members.

Groups. Instructors create groups, add their connected Learners, and chat together. New members cannot see earlier messages.

Online meetings. Group meetings, one-to-one video calls, and session meetings with video, voice, screen sharing, and participant names.

Safety and moderation. Members can block or report others, and admins review reports and can suspend accounts.

7. Technologies to Be Used

Layer	Technology
Front end	HTML5, CSS3 (responsive design), JavaScript, Fetch API for live chat updates, Media Recorder API for voice messages
Back end	Python 3, Flask, Jinja2 templates, Werk Zeug security library for password hashing, JSON for data exchange between browser and server
Database	MySQL, accessed from Flask through a Python MySQL connector
External services	Jitsi Meet (video meetings and screen sharing), Gmail SMTP (verification codes), YouTube embedding (Instructor intro videos)
Development tools	Visual Studio Code, pip, MySQL Workbench, browser developer tools

Why MySQL? HunarHub is meant to run live, with many members using it at the same time. MySQL handles several users reading and writing together far better than a single-file database like SQLite; it supports proper user accounts and backups, and it is available on almost every hosting service. That makes it the safer choice for a platform we plan to keep online.

8. Database Design (Main Tables)

All data is stored in a MySQL database. The main tables and their fields are listed below.

Accounts and profiles:

Table	Fields
Users	id, name, email, password (hash), headline, bio, location, languages, education, photo, improvise, profile_done, suspended
Skills	id, user_id, title, category, level, price, duration, outline
Portfolio	id, user_id, title, description, link
Wants	id, user_id, title
Availability	id, user_id, weekday, start_min, end_min

Community:

Community Posts	id, user_id, group_id (empty for public posts), post_type (tip, question, resource, achievement, opportunity), topic_id, content, image, link, visibility (public, connections, group), created, edited, deleted
Post Comments:	id, post_id, user_id, parent_id (used for replies), body, created, edited, deleted
Reactions:	id, user_id, target_type (post or comment), target_id, reaction (like, helpful, insightful), created
Saved Posts:	user_id, post_id, created
Topics (Topic Follows):	id, name, category user_id, topic_id, created
Notifications:	id, user_id (receiver), actor_id (who caused it), type (request, accepted, message, session, offer, reaction, comment, reply), ref_type, ref_id, message, is_read, created

Connections and messaging:

Connections	id, from_id, to_id, status, notified
Messages	id, from_id, to_id, body, created, is_read, edited, deleted, hide_from, hide_to, file, file_name, file_kind

Sessions, reviews, and Learning Board:

Sessions	id, Instructor_id, learner_id, skill_id, when_ts, hours, note, status, learner_seen, created
Reviews	id, session_id, Instructor_id, learner_id, rating, comment, created
Learning Requests (Wishes)	id, user_id, title, category, details, status, created
Offers	id, wish_id, Instructor_id, message, status, Instructor_seen, created

Groups and meetings:

Study Groups	id, name, description, owner_id, created
Study Members	group_id, user_id, last_read, first_id, joined
Study Messages	id, group_id, from_id, body, kind, deleted, created
Meetings	id, kind, ref_id, peer_id, room, title, started_by, created

Safety and security:

Blocks	id, blocker_id, blocked_id, created
Reports	id, reporter_id, reported_id, message_id, reason, details, snapshot, status, created
Security (Reset Codes, Rate Events)	email, code_hash, expires, attempts; kind, key, created

9. Methodology

We will follow the Software Development Life Cycle with an iterative approach. Each module is built, tested, and improved before the next one is added.

1. **Requirement analysis.** Understanding what learners, Instructors, and administrators need, and studying how existing tutoring and freelancing platforms work.
2. **System design.** Planning the MySQL database structure, the overall architecture, and the user interfaces.

3. **Implementation.** Building the system module with Python, Flask, MySQL, HTML, CSS, and JavaScript.
4. **Testing.** Functional testing of every feature, security testing (wrong passwords, opening pages without permission), and usability testing on desktop and mobile screens.
5. **Deployment.** Putting the system on a web server with HTTPS and a hosted MySQL database so members can use it online.
6. **Maintenance.** Fixing errors, collecting feedback, and adding improvements after launch.

10. Expected Outcomes

When finished, the system should:

- Give Learners and professionals one place to teach and learn skills.
- Make finding the right Instructor easy through search, Smart Match, and Skill Swap.
- Build trust through profiles, portfolios, ratings, and reviews.
- Remove the need for separate chat, scheduling, and meeting apps.
- Help Instructors manage many Learners through groups and group meetings.
- Protect members through secure accounts, blocking, reporting, and admin moderation.
- Show practical use of web development, database design, real-time communication, and web security.

11. Future Work

At this stage, HunarHub is a **learning platform**: it is built as a web application where members teach and learn, and no money is involved. This keeps the first version focused and lets us build a real community before anything else.

Once the web application is live on the internet, the next step is a **mobile application**, so members can use HunarHub as an app on their phones and not only through a browser. We plan to build it on top of the same database, so a member sees the same account, chats, and sessions on both.

After the mobile app, once we have a bigger budget and more resources, our plan is to grow HunarHub into an **earning platform** as well, so that people can make an income from the skills they teach. The steps we have in mind for the earning platform are:

- Online payments, so Instructors can charge for sessions and courses.
- A credit-based system, where members earn credits by teaching and spend them on learning.
- Instructor payouts, earnings, dashboards, and payment history.
- Paid courses and premium group classes.
- Mobile app extras such as push notifications for messages, session reminders, and meeting invites.
- Our own meeting infrastructure in place of the external Jitsi service, if usage grows.

12. Conclusion

HunarHub brings profiles, community feed, search, communication, scheduling, reviews, groups, and online meetings into one secure place, so learners and professionals no longer need to juggle many

separate tools. By connecting learners, instructors, and administrators in one system, it offers a convenient and trustworthy way to share skills and knowledge. The project applies web technologies and database management to a real problem, and it has a clear path to grow later into a mobile application and a platform where people can earn from what they know.